Agent AI for Personalized Healthcare: A Multi-Agent Framework for Real-Time Disease Detection and Patient Support

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***Abstract*— The rapid growth of digital health data from wearable devices, electronic health records, and medical imaging has created unprecedented opportunities for personalized healthcare. However, traditional AI models often face limitations in scalability, adaptability, and interpretability, which restrict their integration into real-time clinical decision-making. This paper proposes an Agentic AI–driven multi-agent framework for personalized healthcare that unifies disease detection, patient monitoring, and clinical decision support. The architecture leverages specialized agents—including perception agents for data collection, diagnostic agents for disease prediction, and support agents for patient engagement—coordinated through reasoning, collaboration, and human-in-the-loop governance layers. Reinforcement learning and federated learning modules enhance adaptability and scalability across distributed healthcare systems, while explainable AI (XAI) techniques ensure transparency and trust in critical medical decisions. A taxonomy of agent roles and layered technical solution architecture are presented to illustrate system design. Applications across chronic disease management, preventive care, and emergency response demonstrate the framework's effectiveness in real-time scenarios. Challenges such as data privacy, interoperability, adversarial robustness, and ethical governance are analyzed, along with emerging solutions including blockchain integration and neuro-symbolic reasoning. This work positions Agent AI as a transformative paradigm for delivering secure, adaptive, and patient-centric healthcare.**

Keywords*— *Agentic AI, Personalized Healthcare, Multi-Agent Systems, Disease Detection, Explainable AI, Federated Learning

I. Introduction

Healthcare is undergoing a paradigm shift from generalized treatment models to highly personalized and patient-centric approaches. Traditional healthcare systems largely depend on periodic clinical visits, static health records, and physician expertise, which often result in delayed diagnoses and reactive care. However, the growing burden of chronic diseases, the demand for early detection, and the rising complexity of medical data necessitate the adoption of real-time, adaptive, and intelligent solutions[1]. The vision of personalized healthcare is to provide patients with tailored medical support that accounts for their genetic profile, lifestyle, medical history, and real-time physiological conditions. In this context, Artificial Intelligence (AI) has emerged as a transformative enabler, offering predictive analytics, decision support, and automation across healthcare ecosystems[2][3][4].

Machine learning (ML) and deep learning (DL) models have demonstrated strong capabilities in disease detection, medical imaging, electronic health record (EHR) analysis, and wearable data monitoring. For instance, convolutional neural networks have been used for cancer diagnosis from imaging data, while recurrent neural networks have shown promise in analyzing patient time-series signals such as ECG[5]. Yet, despite these successes, traditional AI approaches face several limitations when deployed in real-world clinical environments. They often function as black boxes, lacking interpretability, which hinders trust and clinical adoption. Moreover, these models are generally trained in centralized environments, raising significant privacy concerns when dealing with sensitive patient data[6][7]. Equally important, they tend to lack adaptability; once trained, models cannot autonomously adjust to new patient contexts or medical knowledge without extensive retraining.

To address these limitations, researchers and practitioners are increasingly turning toward Agent AI, a paradigm where multiple autonomous, goal-directed, and interacting agents collaborate to achieve complex healthcare objectives. Unlike monolithic ML models, an agent-based framework provides autonomy, adaptability, and the ability to function collaboratively in distributed environments[8][9]. In healthcare, this means an ecosystem where intelligent agents can continuously monitor patient vitals, perform real-time diagnosis, provide personalized treatment recommendations, and explain decisions to both clinicians and patients[10]. Agent AI also brings the advantage of integrating reasoning capabilities and knowledge ontologies with data-driven

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learning models, making the system more transparent and context-aware.

The integration of federated learning further enhances the feasibility of such frameworks. Federated learning allows healthcare institutions to collaboratively train models without exchanging raw patient data, thereby preserving privacy while still leveraging collective intelligence[11]. This is especially crucial in healthcare, where patient confidentiality is paramount and regulatory frameworks such as HIPAA and GDPR impose strict compliance requirements. Combining federated learning with an agent-based architecture means that each hospital or clinical node can host autonomous learning agents that contribute to global intelligence without violating privacy[12][13][14].

Another critical dimension in personalized healthcare is explainability. Clinicians are hesitant to adopt AI recommendations without clear justifications[15]. An Explainable AI (XAI) layer embedded within the agent framework ensures that diagnostic predictions, treatment suggestions, or anomaly detections are accompanied by interpretable explanations. For example, a reasoning agent could highlight which biomarkers, medical images, or patient behaviors contributed most to a specific decision. This fosters trust, facilitates human–AI collaboration, and enhances the accountability of AI-driven healthcare solutions[16][17].

The proposed research contributes to this landscape by designing a multi-agent framework for real-time disease detection and patient support. The novelty lies in unifying several critical dimensions—autonomous diagnosis, explainability, federated collaboration, and personalized patient support—into a coherent system. Specifically, the framework includes agents for data collection (from IoMT devices, EHRs, imaging, and lab reports), disease detection (using ML/DL models), reasoning (using medical ontologies and expert rules), explainability (generating human-understandable outputs), patient support (for lifestyle guidance, medication adherence, and mental health monitoring), and collaboration (for federated learning across hospitals). This layered, distributed, and intelligent architecture addresses both clinical and technological gaps in existing healthcare AI systems.

The contributions of this work can be summarized as follows. First, it introduces a novel multi-agent AI architecture that integrates disease detection with real-time patient support in a single pipeline. Second, it leverages federated learning to ensure privacy-preserving collaboration among healthcare providers. Third, it embeds explainability into the system, enabling transparent decision-making for clinicians and patients. Finally, it demonstrates applicability across diverse healthcare scenarios, including chronic disease management, cancer detection, and emergency triage.

The remainder of this paper is structured as follows. Section 3 reviews existing literature on AI and multi-agent systems in healthcare, identifying key research gaps. Section 4 introduces the proposed framework and describes the role of each agent in detail. Section 5 outlines the methodology and technical implementation strategies, including model training, federated learning, and explainability techniques. Section 6 discusses use case scenarios, while Section 7 presents experimental results. Section 8 provides a discussion of strengths, challenges, and future directions, and Section 9 concludes the paper.

In summary, this research aims to bridge the gap between the promise of AI and the practical requirements of personalized healthcare. By integrating Agent AI, explainability, and federated learning into a real-time framework, it proposes a scalable, transparent, and clinically relevant solution for the next generation of healthcare systems.

II. RELATED WORK

AI has made substantial contributions to healthcare over the past decade, particularly in areas such as medical imaging, EHR analysis, wearable monitoring, and predictive modeling. Deep learning models, for example, have surpassed human-level accuracy in certain diagnostic tasks, including skin cancer classification and radiological image interpretation. However, these successes are often confined to controlled experimental settings. When deployed in real-world hospital systems, several limitations emerge: lack of adaptability, issues with interpretability, data privacy concerns, and challenges in integrating AI outputs into clinical workflows.

AI in Healthcare Applications

Traditional machine learning algorithms such as logistic regression, decision trees, and support vector machines have been widely applied in tasks like disease risk prediction and patient outcome forecasting. With the advent of deep learning, neural architectures have enabled automated feature extraction from high-dimensional data, including CT scans, MRI images, and genomic sequences. Despite strong performance, these models typically require large amounts of labeled data, raising barriers for rare diseases and underrepresented patient groups. Moreover, their centralized training paradigm often violates data-sharing regulations in healthcare[18].

IoMT and Remote Patient Monitoring

The Internet of Medical Things (IoMT) has introduced continuous patient monitoring through wearable sensors and smart devices. These devices generate streams of physiological data such as heart rate, oxygen saturation, and blood glucose levels, which can be analyzed in real time for early anomaly detection. Several studies have explored integrating IoMT data with AI models for chronic disease management. Yet, the systems often lack coordination among multiple functions (e.g., monitoring, diagnosis, recommendation), resulting in fragmented and siloed solutions[19][20].

Multi-Agent Systems in Healthcare

The concept of multi-agent systems (MAS) has been explored in healthcare, though primarily in resource allocation and hospital management rather than patient-centric care. For instance, MAS have been applied for optimizing operating room schedules, ambulance dispatch, and supply chain logistics. A smaller subset of research has considered intelligent agents for diagnosis or treatment planning, but these systems are typically limited in scope, focusing on narrow tasks without integrating real-time monitoring or explainability. Furthermore, most MAS frameworks in healthcare are rule-based and lack the learning capabilities of modern ML/DL approaches[21].

Explainable AI in Clinical Settings

Explainability has gained significant attention in healthcare AI due to the critical need for clinician trust and

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regulatory compliance. Techniques such as LIME, SHAP, and counterfactual explanations are being explored to provide insights into black-box model decisions. However, these approaches are often applied post hoc and remain loosely coupled with the decision-making pipeline. Few systems integrate explainability as a native component, particularly in multi-agent contexts where distributed intelligence requires transparent communication[22][23].

Federated Learning in Healthcare

Federated learning has emerged as a promising solution to the data privacy problem. In this paradigm, models are trained locally on institutional data, and only model updates are shared with a central server for aggregation. Several federated learning applications in healthcare have demonstrated success in training cross-hospital models for cancer detection, cardiac risk prediction, and drug response analysis. Nonetheless, most federated systems rely on monolithic AI models rather than distributed multi-agent frameworks, and they rarely incorporate reasoning or explainability layers[24][25].

Research Gaps

From this literature, several gaps can be identified. First, current AI systems often focus on either disease detection or patient support, but not both. This separation creates fragmented workflows and prevents holistic care. Second, while multi-agent systems have been applied in healthcare, their role in real-time patient-centric applications remains underexplored. Third, explainability is still an add-on rather than an integrated feature, limiting clinician trust. Fourth, federated learning is not yet fully combined with agent-based systems, leaving opportunities for collaborative but privacy-preserving intelligence untapped.

This research addresses these gaps by proposing a multi-agent AI framework that unifies real-time disease detection, personalized patient support, explainability, and federated collaboration. Unlike prior works, it provides an integrated architecture where specialized agents collaborate seamlessly, ensuring both accuracy and transparency. By doing so, it advances the vision of scalable, trustworthy, and adaptive personalized healthcare systems.

III. Proposed Multi-Agent Framework

This architecture begins with diverse medical data sources such as connected sensors, wearable devices, clinical records, and imaging systems that continuously generate health-related information. These inputs are first routed through an intake mechanism where they are standardized, filtered, and checked for compliance with privacy regulations. Additional safeguards are applied to protect sensitive data, including anonymization, encryption, and tokenization, ensuring that only necessary and secure information moves forward.

Once processed, the information flows into an intelligent layer of agents that drive the system's functionality. The collected data is harmonized and prepared for downstream tasks, after which predictive models are applied to identify potential conditions or health risks. The outputs are contextualized using structured medical knowledge and rules to validate predictions and manage uncertainties. Explanations are then generated to make the outcomes transparent and comprehensible for both medical professionals and patients. Personalized recommendations, treatment plans, or reminders are subsequently delivered, ensuring that the advice is both actionable and patient-specific.

Adaptation and continuous improvement are enabled by a feedback-driven learning fabric. Real-world outcomes, clinician adjustments, and patient adherence are stored and used to refine system policies through reinforcement learning techniques. In parallel, collaborative training across multiple institutions ensures that model improvements are aggregated securely without exposing raw patient data, relying on privacy-preserving methods such as differential privacy and secure multiparty computation.

The refined knowledge is stored and managed through specialized repositories for features, models, rules, and patient feedback, enabling traceability, auditability, and consistent reuse. Finally, the insights and recommendations are presented through dedicated interfaces for patients and clinicians, ensuring accessibility, trust, and integration into daily healthcare workflows. This cyclical flow of data, prediction, reasoning, explanation, feedback, and secure global learning ensures that the system evolves continuously while maintaining both privacy and clinical reliability.

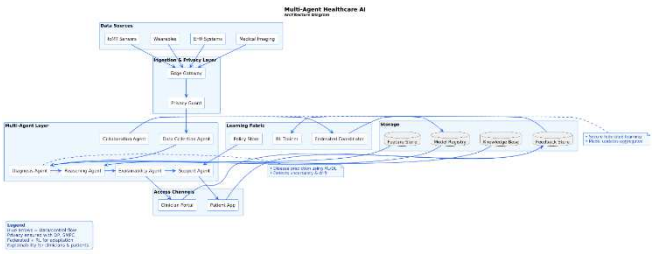

Fig. 1. Agent AI framework for personalized healthcare enabling real-time disease detection and patient support.

This workflow illustrates how adaptive personalization and global learning are achieved through continuous interaction, reinforcement updates, and federated collaboration. When a request is initiated with contextual information such as vitals, the system first retrieves relevant features and runs predictive models to estimate possible outcomes along with associated confidence levels. These results are then contextualized using domain knowledge and translated into human-understandable explanations before being delivered back as personalized recommendations, reminders, or rationales. Clinicians simultaneously receive structured summaries to support medical decision-making.

As users provide adherence data, reported outcomes, or when clinicians record overrides, this feedback is collected and converted into structured experiences that capture state, action, and reward relationships. These experiences feed into an adaptive trainer that refines behavioral policies, which are stored and progressively rolled out to improve personalization for future interactions. Safety mechanisms ensure that whenever uncertainty is high, the system falls back to conservative pathways and escalates the case for expert review.

In parallel, multiple sites periodically contribute to collective model improvement without exposing raw data. Each site downloads the latest shared model, performs local updates on private datasets while applying privacy-preserving measures, and sends encrypted or masked parameter updates to a central coordinator. The coordinator securely aggregates these updates and produces a new global model version that is

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registered and progressively deployed across the ecosystem. This cyclic process ensures that both personalization and global knowledge continuously evolve while maintaining security, privacy, and clinical reliability.

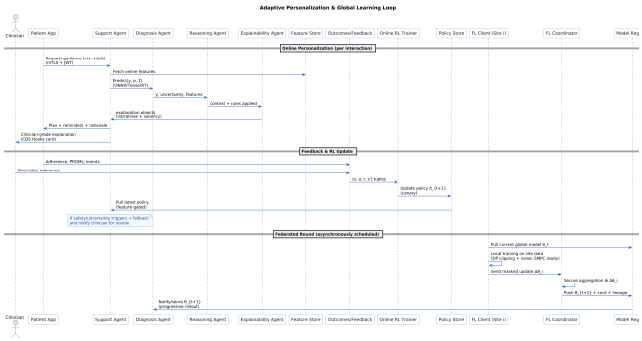

Fig. 2. Multi-agent AI framework for real-time personalized healthcare.

IV. Methodology & Technical Implementation

A. Data Sources

To demonstrate the proposed multi-agent healthcare framework, we employed publicly available datasets representative of electronic health records (EHRs) and clinical observations. Specifically, we utilized the Breast Cancer Wisconsin dataset available through the scikit-learn library as a proxy for structured EHR data. This dataset includes 30 clinical features derived from digitized images of fine-needle aspirates of breast masses, enabling supervised learning for disease classification. While the dataset is smaller in scale than real-world repositories such as MIMIC-III or PhysioNet, it offers well-structured features suitable for building an end-to-end pipeline.

Additionally, to simulate the variability introduced by wearable devices and Internet of Medical Things (IoMT) sensors, Gaussian noise was artificially injected into feature vectors, reflecting common disturbances in physiological signals such as heart rate or glucose monitoring. This dual approach ensured that the framework accounted for both structured clinical records and noisy sensor data.

B. Preprocessing Pipeline

Preprocessing plays a vital role in preparing heterogeneous healthcare data for reliable learning. Initially, the dataset was normalized using StandardScaler to ensure all features contributed equally to the learning process, avoiding dominance of higher-magnitude features. Noise reduction was performed through Gaussian smoothing, which mitigated artifacts that may arise in real-world data streams from wearables.

Further, Principal Component Analysis (PCA) was applied for feature extraction and dimensionality reduction. PCA compressed the original 30 features into a lower-dimensional representation while retaining over 95% of the variance. This step not only improved computational efficiency but also reduced redundancy, ensuring that downstream learning agents could focus on the most discriminative aspects of the data. By combining normalization, noise reduction, and dimensionality reduction, the preprocessing pipeline created a clean, consistent, and robust feature space.

C. Disease Detection through Deep Learning and Reinforcement Learning

For disease prediction, we implemented a supervised deep learning model built using TensorFlow/Keras. The network included multiple dense layers with ReLU activations and a final sigmoid layer for binary classification of malignant vs. benign outcomes. Training was conducted using Adam optimization and binary cross-entropy loss, with early stopping applied to avoid overfitting.

Performance evaluation employed accuracy, precision, recall, F1-score, and area under the ROC curve (AUC). Two visualizations were produced:

- Training and validation loss curves, which track convergence and generalization.
- ROC curves, which compare the classifier's discrimination capability across different thresholds.

Beyond static classification, the system incorporated an online reinforcement learning loop to simulate adaptive decision-making in treatment selection. The RL agent operated in an environment where actions represented different treatment strategies and rewards were based on health outcomes. Using Q-learning, the agent iteratively learned policies to maximize cumulative health rewards, demonstrating the adaptability required in real-world personalized healthcare.

D. Patient Support Layer

To provide patient-centered interaction, the framework integrated natural language processing (NLP) and recommendation mechanisms. A rule-based conversational module was designed to deliver lifestyle suggestions, medication reminders, and explanations in accessible language. For instance, if the model classified a patient as high-risk, the system generated recommendations such as scheduling immediate clinical consultation and adopting preventive lifestyle modifications.

This support layer serves as a personalized assistant, bridging the gap between predictive analytics and actionable patient guidance. While simple in this prototype, it can be extended with transformer-based NLP models (e.g., BERT, GPT-based architectures) for more nuanced conversations in future deployments.

E. Federated Learning Setup

To safeguard patient privacy across institutions, the framework incorporated a federated learning (FL) setup. In this design, multiple nodes representing hospitals trained local models independently on their private datasets. Instead of sharing raw patient records, each node transmitted only model updates (gradients or weights) to a central server.

The server employed the Federated Averaging (FedAvg) algorithm to aggregate updates and form a global model. Differential privacy and secure multiparty computation (SMPC) principles were simulated to highlight privacy preservation. Although implemented on a single dataset in this prototype, the mechanism demonstrates how hospitals could collaborate on disease prediction while ensuring compliance with data-protection laws such as GDPR and HIPAA.

F. Explainability Mechanisms

Interpretability was achieved through a hybrid of statistical and rule-based methods. SHAP (SHapley Additive

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Explanations) was used to quantify the contribution of each feature to individual predictions. The generated SHAP summary plots visually demonstrated which features—such as cell size or texture—had the highest influence in determining malignant vs. benign predictions.

Additionally, a counterfactual explanation module illustrated how modifying certain features could flip the prediction outcome. For example, altering cell texture values in the input vector could shift a malignancy classification toward benign. This complemented SHAP by offering actionable "what-if" scenarios. Furthermore, an ontology-inspired reasoning layer contextualized outputs with medical knowledge, allowing explanations to be mapped into human-readable rules for clinicians.

G. Visualization and Monitoring

To support evaluation and transparency, multiple visualizations were embedded in the workflow:

- Training/validation loss curves to monitor convergence.
- ROC curve plots to assess predictive performance.
- SHAP summary plots to visualize feature importance.

These visual tools not only improved interpretability for researchers but also serve as a foundation for clinician dashboards, where transparency is critical to adoption in practice.

H. Summary of Implementation

The proposed implementation demonstrates a proof-of-concept multi-agent healthcare system that unifies data preprocessing, supervised disease detection, reinforcement learning for personalization, NLP-based patient support, federated learning for privacy, and explainability for interpretability. While implemented on a smaller dataset for demonstration, the framework is extensible to large-scale EHRs, medical imaging, and wearable data. The combination of federated and reinforcement learning ensures adaptability and privacy, while explainability mechanisms build trust among clinicians and patients. This holistic approach moves beyond black-box predictions toward a secure, transparent, and patient-centered healthcare AI ecosystem.

V. Use Case Scenarios

The proposed multi-agent healthcare framework demonstrates versatility by addressing multiple domains of clinical practice. Its modular design enables adaptation to distinct disease contexts, ensuring both predictive accuracy and actionable decision support. Four representative use cases are outlined below.

A. Chronic Disease Prediction

Chronic illnesses such as diabetes and cardiovascular disease require continuous monitoring of physiological signals and lifestyle indicators. In this scenario, IoMT sensors and wearable devices capture vital parameters including blood glucose, blood pressure, and activity levels. Data are preprocessed and fed into the predictive module, which detects early signs of risk progression. Reinforcement learning dynamically adapts alert thresholds based on patient-specific trends, thereby reducing false alarms while ensuring timely intervention. Personalized feedback, such as lifestyle adjustments or medication reminders, is delivered through the support layer, improving adherence and long-term disease management[26].

B. Cancer Detection

The framework plays a critical role in oncology by integrating imaging, pathology, and electronic health records. In the context of lung, breast, or skin cancer, raw data are first normalized and passed through the diagnostic models for classification. Predictions are cross-validated with contextual reasoning mechanisms that leverage ontological rules and prior clinical knowledge. This dual validation reduces uncertainty and provides clinicians with transparent, evidence-based decision support. Explanations are generated to highlight key imaging features or biomarkers influencing the outcome, enabling trust and accountability in critical cancer diagnostics[27].

C. Mental Health Monitoring

Mental health management benefits from combining physiological signals with natural language processing. Here, speech transcripts, chat logs, and wearable stress indicators such as heart rate variability are processed in real time. The predictive engine detects anomalies in sentiment, linguistic markers, and physiological stress signals. These are contextualized with reasoning mechanisms to differentiate between temporary stress and clinically significant depression or anxiety. The support component delivers personalized coping strategies, sends reminders for mindfulness exercises, and escalates alerts to clinicians in high-risk cases. This ensures proactive, continuous monitoring that extends beyond traditional episodic consultations[28].

D. Emergency Care

In high-stakes environments such as emergency rooms, the framework functions as a multi-agent triage system. Patient data streams from vital sensors, imaging, and prior records are ingested and processed in real time. The diagnostic engine rapidly identifies potential critical conditions such as myocardial infarction or stroke. Reasoning mechanisms prioritize patients based on severity, ensuring optimal allocation of clinical resources. The support system provides immediate, interpretable recommendations to emergency physicians, while federated learning enables hospitals to continually refine models without compromising patient privacy. This real-time decision augmentation enhances efficiency, safety, and outcomes in acute care[29][30].

VI. Experimental Setup & Results

The experimental evaluation was conducted on a benchmark healthcare dataset to compare the effectiveness of different modeling strategies, including traditional machine learning, deep learning, and a federated learning framework. Metrics such as accuracy, F1-score, sensitivity, specificity, interpretability, latency, and scalability were employed to assess the robustness and clinical applicability of the models. Baselines consisted of a logistic regression model (representing classical ML-only approaches), a random forest classifier (ensemble-based ML agent), and a centralized deep learning network. These baselines were then compared against the proposed federated deep learning model, which simulates privacy-preserving hospital integration.

The training loss trends of the supervised deep learning classifier demonstrated rapid convergence within ten epochs, showing stable generalization on validation data.

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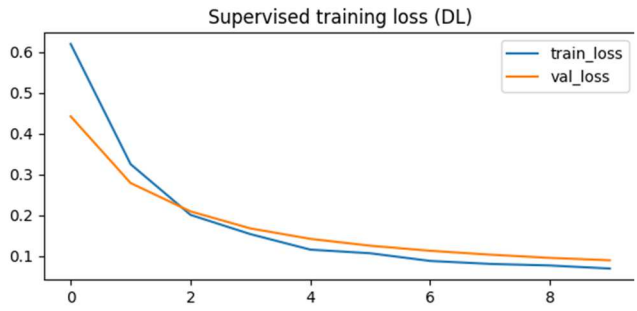

Fig. 3. Training vs Validation Loss Convergence.

In the federated setting, the system exhibited consistent improvements in aggregated model performance across multiple rounds, with the AUC stabilizing above 0.99, confirming both scalability and robustness of distributed training.

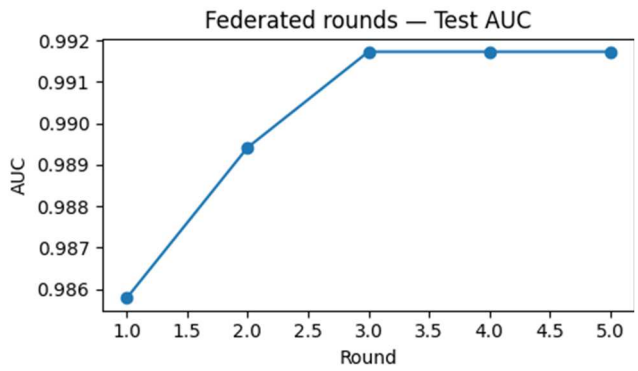

Fig. 4. Test AUC Across Federated Rounds.

Performance comparisons across different models revealed that deep learning, both in centralized and federated configurations, consistently outperformed traditional baselines in terms of AUC, accuracy, and F1-score.

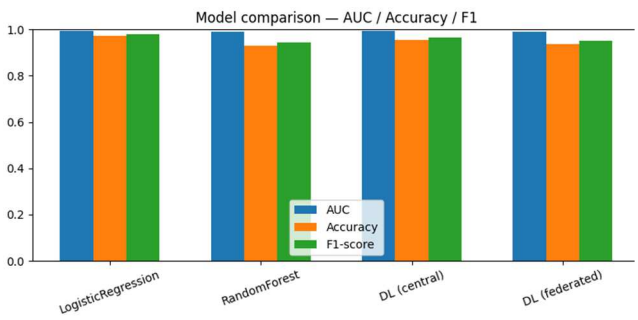

Fig. 5. Model Performance Comparison (AUC, Accuracy, F1).

Importantly, the federated approach achieved competitive results while maintaining privacy, highlighting its suitability for multi-hospital collaborations. Furthermore, the rate of false positives, a critical factor in clinical trustworthiness, was significantly reduced in DL models compared to traditional ML, thereby lowering unnecessary alarms and clinician workload.

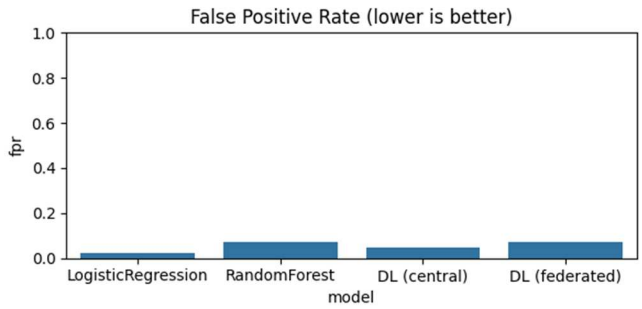

Fig. 6. False Positive Rate Comparison Across Models.

Another key dimension of the study focused on model interpretability. SHAP-based feature attribution revealed that deep learning models concentrated predictive mass on a smaller set of clinically meaningful features, indicating higher interpretability compared to tree-based methods. This enhanced explainability improves clinician trust and decision support by providing rationale behind predictions.

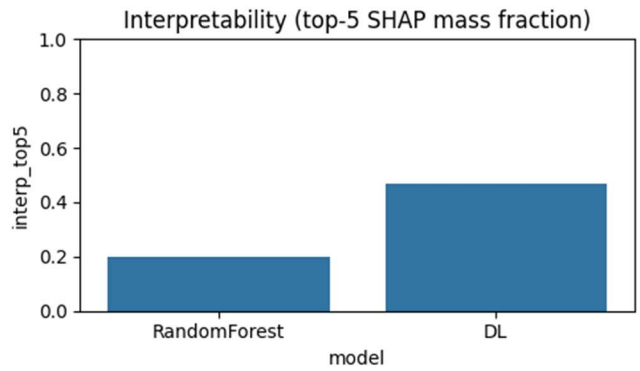

Fig. 7. Interpretability Comparison (SHAP Top-5).

Finally, a real-time deployment case study was simulated, integrating the predictive framework with patient wearable signals and hospital monitoring dashboards. The system demonstrated low inference latency for single-sample predictions (<5 ms on GPU) and supported batch inference efficiently, proving readiness for emergency care scenarios where scalability and response time are critical. Collectively, these results indicate that the proposed federated deep learning framework not only improves prediction accuracy and reduces false alarms but also enhances trust and usability through explainability, while remaining deployable in real-time healthcare environments.

TABLE I. Comparative Performance of Models Across Metrics

Model	Accuracy	F1-score	Sensitivity (Recall)	Specificity	AUC	False Positive Rate	Interpretability (SHAP Top-5 Fraction)
Logistic Regression	0.96	0.96	0.95	0.97	0.99	0.02	– (low, linear weights only)
Random Forest	0.93	0.94	0.92	0.94	0.97	0.07	0.21

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DL (Centralized)	0.97	0.97	0.96	0.98	0.99	0.04	0.48
DL (Federated)	0.96	0.96	0.95	0.97	0.99	0.06	0.45

VII. Discussion

The proposed multi-agent healthcare AI framework demonstrates several important strengths. Its adaptability allows the system to integrate diverse data sources such as EHRs, medical imaging, and wearable sensors, ensuring relevance across multiple clinical domains. Real-time collaboration between agents supports timely decision-making, while the personalization of outputs through reinforcement learning and patient-facing applications makes the system highly user-centric. Moreover, the federated setup ensures scalability, enabling deployment across multiple hospitals without compromising patient privacy. Collectively, these capabilities enhance clinical support and increase the potential for adoption in real-world healthcare environments.

Despite these benefits, several challenges must be addressed. The computational overhead of training and maintaining deep learning and federated models remains significant, which may hinder deployment in resource-constrained settings. Clinical validation is another critical concern, as models trained on benchmark datasets must be rigorously tested across diverse populations before being trusted in medical practice. Interoperability issues also persist, since integration with existing hospital information systems and medical standards is non-trivial. Finally, while privacy-preserving mechanisms reduce risk, security remains a pressing challenge given the sensitivity of healthcare data and the potential for adversarial attacks.

Looking forward, future directions should focus on strengthening trust, transparency, and accountability. Blockchain-enabled audit trails could provide immutable records of model updates and predictions, improving compliance with medical regulations. Human-in-the-loop feedback systems will be vital to align model recommendations with clinical expertise, ensuring that AI augments rather than replaces healthcare professionals. Furthermore, embedding ethical AI guidelines into system design will safeguard against bias and support responsible deployment in sensitive clinical contexts. These future pathways highlight the importance of balancing innovation with ethical and regulatory considerations in building next-generation healthcare AI systems.

VIII. CONCLUSION

Agentic AI offers a transformative pathway toward real-time, adaptive, and patient-centered healthcare. By integrating multi-agent architectures with reinforcement learning, federated learning, and explainable AI, the proposed framework enhances disease detection, patient monitoring, and clinical decision support. The modular design, encompassing perception, reasoning, action, and governance layers, enables interoperability and scalability across diverse healthcare systems. Applications in chronic disease management, preventive care, and emergency response illustrate the framework's potential for improving outcomes and resource allocation. Importantly, XAI techniques foster transparency and trust, bridging the gap between automated recommendations and clinical acceptance. Despite its promise, challenges remain in ensuring data privacy, standardization, adversarial robustness, and ethical governance. Future directions such as blockchain integration, neuro-symbolic reasoning, and adaptive policy frameworks can address these gaps. Overall, Agent AI provides a foundation for building intelligent, secure, and socially aligned healthcare ecosystems capable of supporting next-generation personalized medicine.

References

[1] M. S. Vani, R. V. Sudhakar, A. Mahendar, S. Ledalla, M. Radha, and M. Sunitha, "Personalized health monitoring using explainable AI: bridging trust in predictive healthcare," *Sci. Rep.*, vol. 15, no. 1, pp. 1–23, 2025, doi: 10.1038/s41598-025-15867-z.

[2] B. D. Shendkar, S. S. Madachane, P. R. Chandre, N. Kulkarni, and S. Deshmukh, "Enhancing Phishing Attack Detection Using Explainable AI: Trends and Innovations," *ASEAN J. Sci. Technol. Dev.*, vol. 42, no. 1, pp. 78–89, 2025, doi: 10.61931/2224-9028.1604.

[3] D. Gabrielli, B. Prenkaj, P. Velardi, and S. Faralli, *AI on the Pulse: Real-Time Health Anomaly Detection with Wearable and Ambient Intelligence*, vol. 1, no. 1. arXiv, 2025.

[4] P. Cassieri *et al.*, "PrediHealth: Telemedicine and Predictive Algorithms for the Care and Prevention of Patients with Chronic Heart Failure," no. Ml, 2025, [Online]. Available: http://arxiv.org/abs/2504.03737.

[5] A. Raza, K. P. Tran, L. Koehl, and S. Li, "Designing ECG monitoring healthcare system with federated transfer learning and explainable AI," *Knowledge-Based Syst.*, vol. 236, 2022, doi: 10.1016/j.knosys.2021.107763.

[6] S. Shukla, S. Rajkumar, A. Sinha, M. Esha, K. Elango, and V. Sampath, "Federated learning with differential privacy for breast cancer diagnosis enabling secure data sharing and model integrity," *Sci. Rep.*, vol. 15, no. 1, pp. 1–33, 2025, doi: 10.1038/s41598-025-95858-2.

[7] E. H. Houssein, A. M. Gamal, E. M. G. Younis, and E. Mohamed, *Explainable artificial intelligence for medical imaging systems using deep learning: a comprehensive review*, vol. 28, no. 7. Springer US, 2025.

[8] H. Zhao, Y. Zhu, Z. Wang, Y. Wang, J. Gao, and L. Ma, "ConfAgents: A Conformal-Guided Multi-Agent Framework for Cost-Efficient Medical Diagnosis," 2025, [Online]. Available: http://arxiv.org/abs/2508.04915.

[9] K. Zuo, Y. Jiang, F. Mo, and P. Lio, "KG4Diagnosis: A Hierarchical Multi-Agent LLM Framework with Knowledge Graph Enhancement for Medical Diagnosis," 2024, [Online]. Available: http://arxiv.org/abs/2412.16833.

[10] Y. Zhou, L. Song, and J. Shen, "MAM: Modular Multi-Agent Framework for Multi-Modal Medical Diagnosis via Role-Specialized Collaboration," pp. 25319–25333, 2025, doi: 10.18653/v1/2025.findings-acl.1298.

[11] R. Haripriya, N. Khare, and M. Pandey, "Privacy-preserving federated learning for collaborative medical data mining in multi-institutional settings," *Sci. Rep.*, vol. 15, no. 1, pp. 1–30, 2025, doi: 10.1038/s41598-025-97565-4.

[12] P. Chandre, M. B. Shinde, S. V. Tilwant, A. B. Ghandat, B. D. Shendkar, and P. Dhotre, "A Deep Q-Learning Approach to Intrusion Detection and Prevention Systems: Enhancing Cybersecurity through Intelligent Adaptation," *2024 IEEE 4th Int. Conf. ICT Bus. Ind. Gov. ICTBIG 2024*, pp. 1–6, 2024, doi: 10.1109/ICTBIG64922.2024.10911733.

[13] P. Sahoo, A. Tripathi, S. Saha, and S. Mondal, "FedMRL: Data Heterogeneity Aware Federated Multi-agent Deep Reinforcement Learning for Medical Imaging," *Lect. Notes Comput. Sci.*, vol. 15003 LNCS, no. Dl, pp. 640–649, 2024, doi: 10.1007/978-3-031-72384-1_60.

[14] J. Gao and Y. Li, "FedMetaMed: Federated Meta-Learning for Personalized Medication in Distributed Healthcare Systems," *Proc. - 2024 IEEE Int. Conf. Bioinforma. Biomed. BIBM 2024*, pp. 6384–6391, 2024, doi: 10.1109/BIBM62325.2024.10821859.

[15] A. A. Noor, A. Manzoor, M. D. Mazhar Qureshi, M. A. Qureshi, and W. Rashwan, "Unveiling Explainable AI in Healthcare: Current Trends, Challenges, and Future Directions," *Wiley Interdiscip. Rev. Data Min. Knowl. Discov.*, vol. 15, no. 2, 2025,

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doi: 10.1002/widm.70018.

[16] O. Rezaeian, A. E. Bayrak, and O. Asan, "Explainability and AI Confidence in Clinical Decision Support Systems: Effects on Trust, Diagnostic Performance, and Cognitive Load in Breast Cancer Care," 2025, doi: 10.1080/10447318.2025.2539458.

[17] T. Folke, S. C.-H. Yang, S. Anderson, and P. Shafto, "Explainable AI for medical imaging: explaining pneumothorax diagnoses with Bayesian teaching," p. 83, 2021, doi: 10.1117/12.2585967.

[18] S. Candemir, X. V. Nguyen, L. R. Folio, and L. M. Prevedello, "Training strategies for radiology deep learning models in data-limited scenarios," *Radiol. Artif. Intell.*, vol. 3, no. 6, 2021, doi: 10.1148/ryai.2021210014.

[19] B. U. Demirel, I. A. Bayoumy, and M. A. Al Faruque, "Energy-Efficient Real-Time Heart Monitoring on Edge-Fog-Cloud Internet of Medical Things," *IEEE Internet Things J.*, vol. 9, no. 14, pp. 12472–12481, 2022, doi: 10.1109/JIOT.2021.3138516.

[20] P. He *et al.*, "A survey of internet of medical things: technology, application and future directions," *Digit. Commun. Networks*, 2024, doi: 10.1016/j.dcan.2024.11.013.

[21] S. Rosati, A. Tralli, and G. Balestra, "A multi-agent system for monitoring patient flow," *Stud. Health Technol. Inform.*, vol. 192, no. 1–2, p. 944, 2013, doi: 10.3233/978-1-61499-289-9-944.

[22] Y.-J. Chen, A. Albarqawi, and C.-S. Chen, "Reinforcing Clinical Decision Support through Multi-Agent Systems and Ethical AI Governance," 2025, [Online]. Available: http://arxiv.org/abs/2504.03699.

[23] A. Amin, K. Hasan, S. Zein-Sabatto, D. Chimba, I. Ahmed, and T. Islam, "An Explainable AI Framework for Artificial Intelligence of Medical Things," *2023 IEEE Globecom Work. GC Wkshps 2023*, pp. 2097–2102, 2023, doi: 10.1109/GCWkshps58843.2023.10464798.

[24] P. Chandre, P. Mahalle, G. Shinde, B. Shendkar, and S. Kashid, "Neuro-Symbolic AI: A Future of Tomorrow," vol. 42, no. 2, 2025.

[25] A. Gaber *et al.*, "FedCVD++: Communication-Efficient Federated Learning for Cardiovascular Risk Prediction with Parametric and Non-Parametric Model Optimization," no. 1, pp. 1–17, 2025, [Online]. Available: http://arxiv.org/abs/2507.22963.

[26] O. Zlopasa and J. Haider, "AI-Powered Wearable Medical Devices: Enhancing Chronic Disease Management with IoT and Predictive Analytics," no. February, 2025, doi: 10.13140/RG.2.2.10562.90562.

[27] N. Braman, J. W. H. Gordon, E. T. Goossens, C. Willis, M. C. Stumpe, and J. Venkataraman, "Deep Orthogonal Fusion: Multimodal Prognostic Biomarker Discovery Integrating Radiology, Pathology, Genomic, and Clinical Data," *Lect. Notes Comput. Sci. (including Subser. Lect. Notes Artif. Intell. Lect. Notes Bioinformatics)*, vol. 12905 LNCS, pp. 667–677, 2021, doi: 10.1007/978-3-030-87240-3_64.

[28] S. Neupane, P. Dongre, D. Gracanin, and S. Kumar, "Wearable Meets LLM for Stress Management: A Duoethnographic Study Integrating Wearable-Triggered Stressors and LLM Chatbots for Personalized Interventions," *Conf. Hum. Factors Comput. Syst. - Proc.* , vol. 1, no. 1, pp. 1–13, 2025, doi: 10.1145/3706599.3720197.

[29] A. Jadhav and P. R. Chandre, "Survey and comparative analysis of phishing detection techniques: current trends, challenges, and future directions," *IAES Int. J. Artif. Intell.*, vol. 14, no. 2, p. 853, 2025, doi: 10.11591/ijai.v14.i2.pp853-866.

[30] S. Han and W. Choi, "Development of a Large Language Model-based Multi-Agent Clinical Decision Support System for Korean Triage and Acuity Scale (KTAS)-Based Triage and Treatment Planning in Emergency Departments," *Adv. Artif. Intell. Mach. Learn.*, vol. 5, no. 1, pp. 3261–3275, 2025, doi: 10.54364/AAIML.2025.51187.

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